

Tibshirani Homework 1

Simulated Questions Only

Afraz Arif

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Introduction

This document contains only the simulated / coding-based questions that were completed in Python: Q6, Q7, Q10, and Q17. For Q6,

Q5 Recap

For the random walk

$$x_t = \delta + x_{t-1} + w_t, \quad w_t \sim N(0, \sigma^2),$$

taking first differences gives

$$\Delta x_t = x_t - x_{t-1} = \delta + w_t.$$

Therefore, the differenced observations are i.i.d. Gaussian with mean δ . To test

$$H_0 : \delta = 0,$$

I use the standard one-sample t -statistic applied to the differenced series:

$$T = \frac{\bar{\Delta x}}{s_{\Delta x} / \sqrt{n}},$$

where $\bar{\Delta x}$ is the sample mean of the differences and $s_{\Delta x}$ is the sample standard deviation. Under the null hypothesis, this statistic has a t_{n-1} distribution.

Q6: Simulate a random walk and compute the test statistic

I simulated a random walk of length 200 with no drift, and then repeated the experiment with a large nonzero drift.

Using the Python code, I obtained:

- No drift ($\delta = 0$): $t \approx -0.4883$
- Large drift ($\delta = 3$): $t \approx 41.8932$

These results behave as expected. In the no-drift case, the statistic is close to zero, which is consistent with the null hypothesis. In the large-drift case, the statistic is very large in magnitude, giving overwhelming evidence against $H_0 : \delta = 0$.

Q7: 50 random walks with nonzero drift

I simulated 50 random walks of length 200 with nonzero drift, plotted all of them on the same figure, and overlaid:

- the sample mean $\hat{\mu}_t$ across the 50 paths,
- the bands $\hat{\mu}_t \pm \hat{\sigma}_t$.

The plot shows two key features:

1. The sample mean increases over time, which reflects the positive drift.
2. The spread of the sample paths also increases over time, which shows that the variance grows with t .

This agrees with the theory for a random walk with drift: the mean grows linearly in time, and the variance also increases over time, so the process is not stationary.

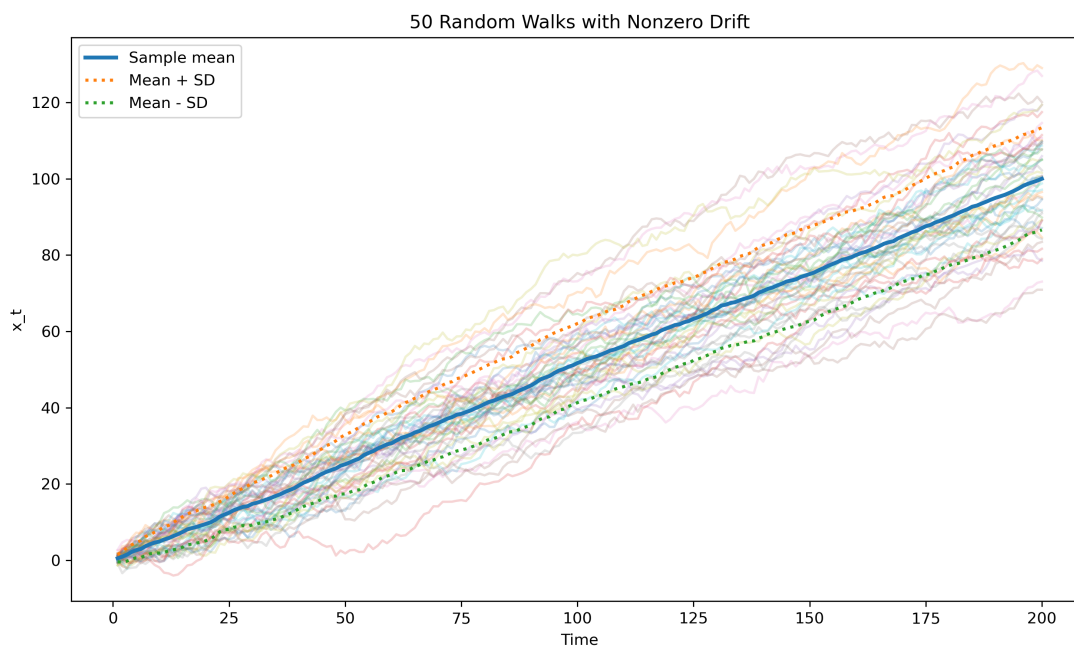


Figure 1: Fifty simulated random walks with nonzero drift, along with the sample mean and the mean $\pm$ one sample standard deviation.

Q10: Simulate the processes from Q8 and Q9

The process is

$$x_t = w_t w_{t-1}.$$

Q8 case: $w_t \sim N(0, \sigma^2)$

From the theoretical calculations:

$$\mathbb{E}[x_t] = 0, \quad \text{Var}(x_t) = \sigma^4.$$

With $\sigma = 1$, this becomes:

$$\mathbb{E}[x_t] = 0, \quad \text{Var}(x_t) = 1.$$

The population autocorrelation function is:

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0, & h \neq 0. \end{cases}$$

Q9 case: $w_t \sim N(\mu, \sigma^2)$ **with** $\mu \neq 0$

From the theoretical calculations:

$$\mathbb{E}[x_t] = \mu^2, \quad \text{Var}(x_t) = \sigma^4 + 2\mu^2\sigma^2.$$

With $\mu = 2$ and $\sigma = 1$, this becomes:

$$\mathbb{E}[x_t] = 4, \quad \text{Var}(x_t) = 9.$$

In this case the autocorrelation is nonzero at lag 1:

$$\rho(1) = \rho(-1) = \frac{\mu^2\sigma^2}{\sigma^4 + 2\mu^2\sigma^2}.$$

For $\mu = 2$ and $\sigma = 1$, this gives

$$\rho(1) = \rho(-1) = \frac{4}{9}.$$

Simulation summary

From the Python simulation:

- For the Q8 process, the sample mean was close to 0 and the sample variance was close to 1.
- For the Q9 process, the sample mean was close to 4 and the sample variance was close to 9.
- The sample ACF for Q8 was approximately 0 away from lag 0.
- The sample ACF for Q9 showed a visible lag-1 dependence, as predicted by theory.

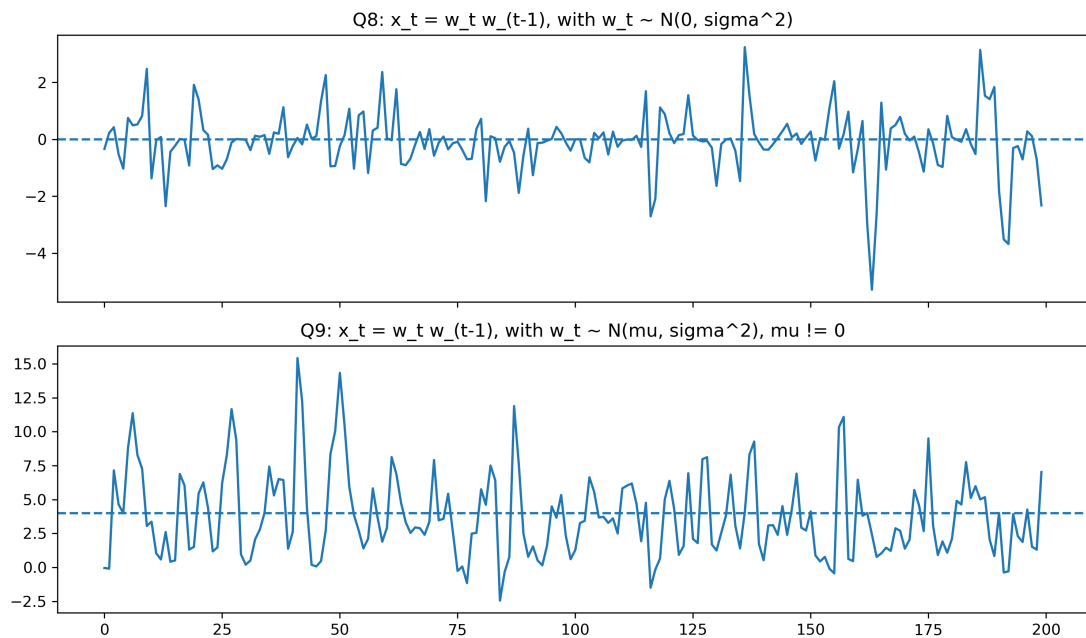


Figure 2: Simulated time series for the Q8 and Q9 processes.

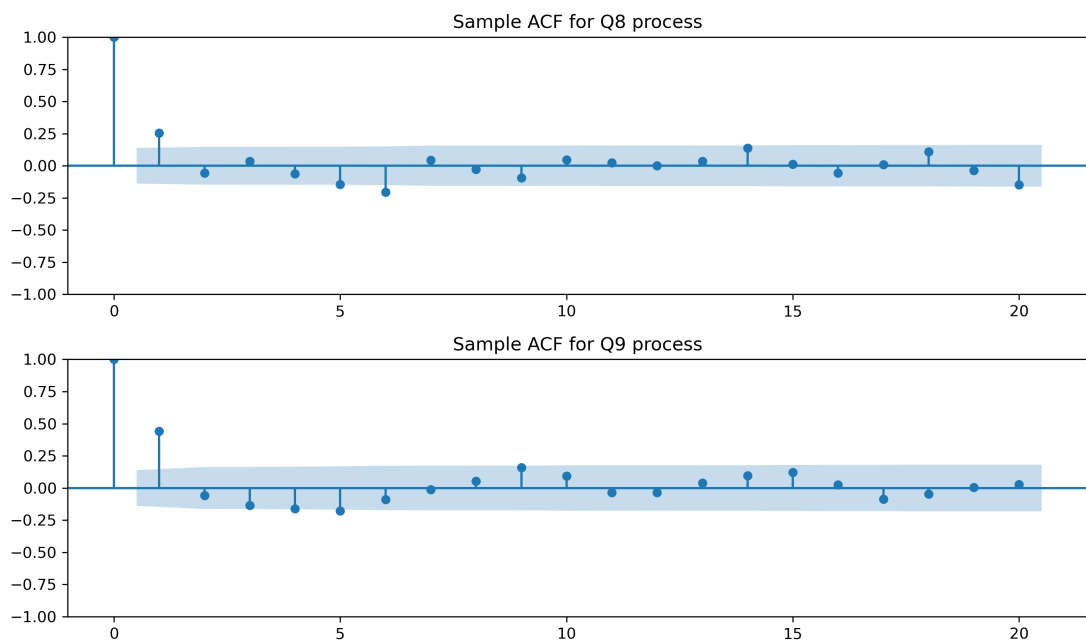


Figure 3: Sample ACFs for the Q8 and Q9 simulated processes.

Q17: Cross-correlation between COVID-19 cases and deaths

For Q17, I computed the sample cross-correlation between daily COVID-19 cases and daily COVID-19 deaths for the following states:

- Florida

- Georgia
- New York
- Pennsylvania
- Texas

I also plotted the case and death signals together after scaling them to the same range.

The main qualitative finding is that the strongest positive cross-correlation typically occurs at a positive lag, which suggests that deaths tend to follow cases after a delay. This is also consistent with the visual signal plots, where the deaths curve often looks like a delayed and smoother version of the cases curve.

There are differences across states in the sharpness and exact location of the peak cross-correlation, but the overall pattern is broadly similar.

Florida

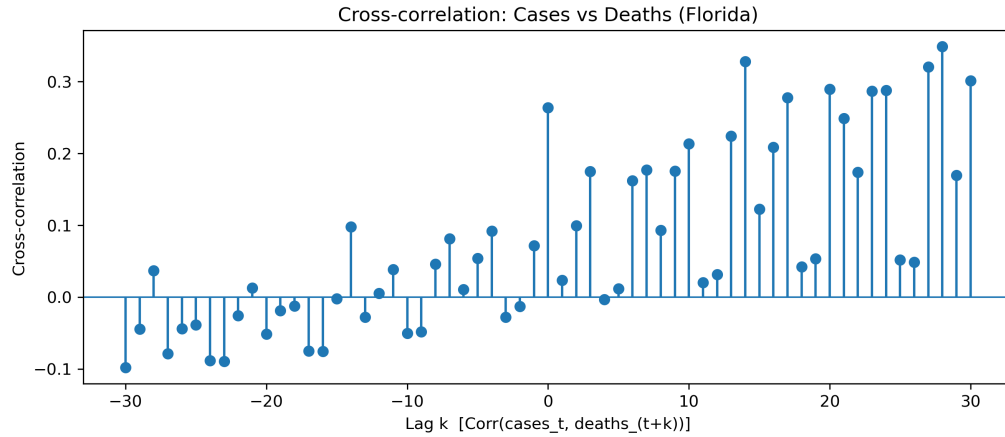


Figure 4: Cross-correlation between cases and deaths for Florida.

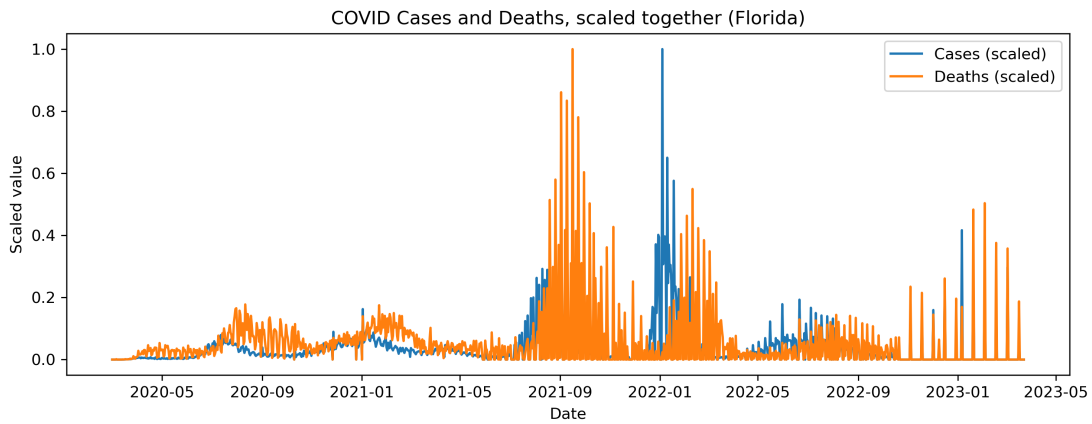


Figure 5: Scaled daily cases and deaths for Florida.

Georgia

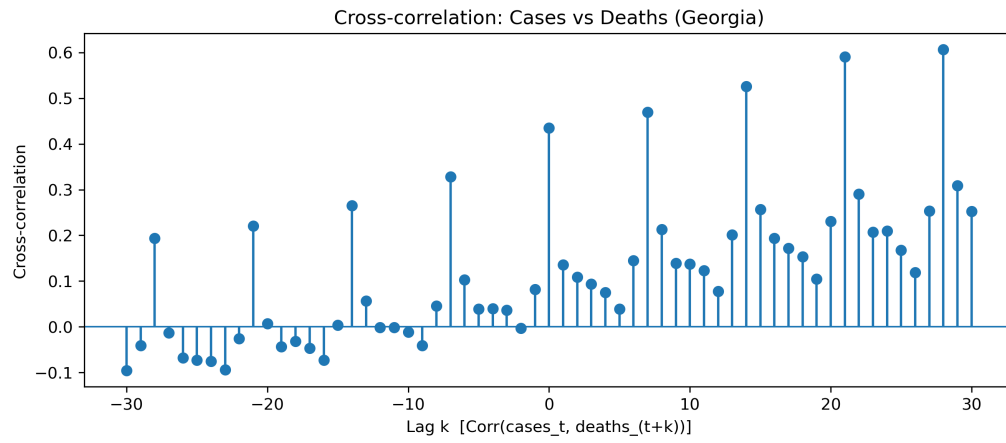


Figure 6: Cross-correlation between cases and deaths for Georgia.

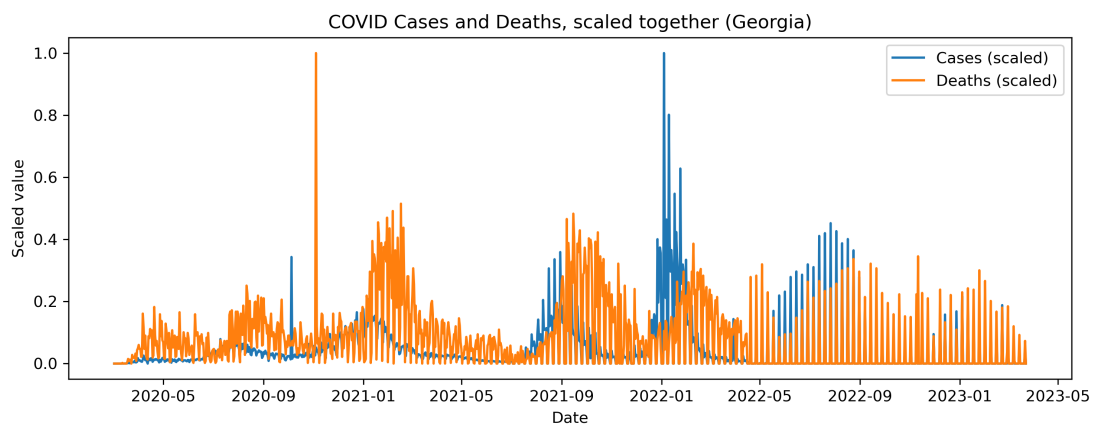


Figure 7: Scaled daily cases and deaths for Georgia.

New York

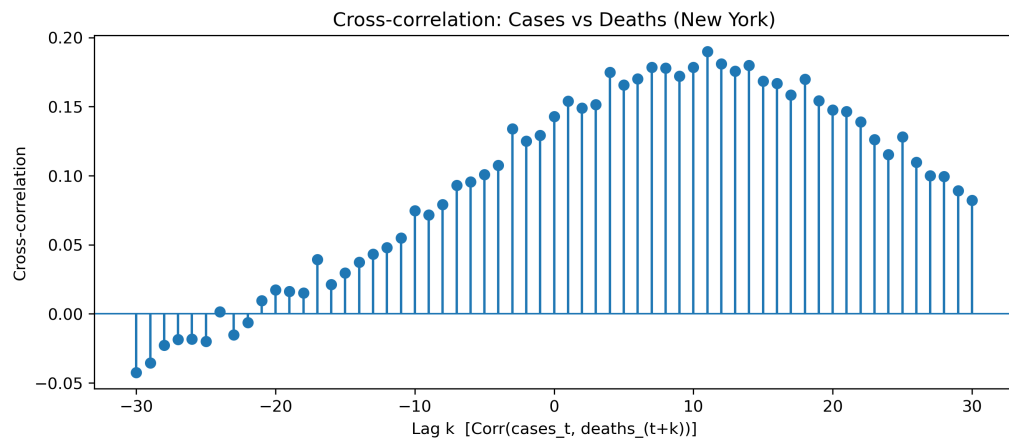


Figure 8: Cross-correlation between cases and deaths for New York.

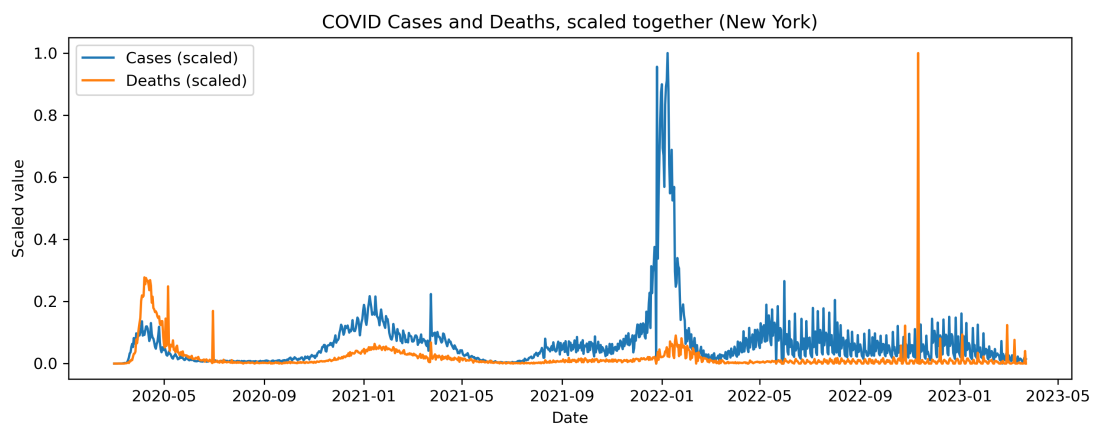


Figure 9: Scaled daily cases and deaths for New York.

Pennsylvania

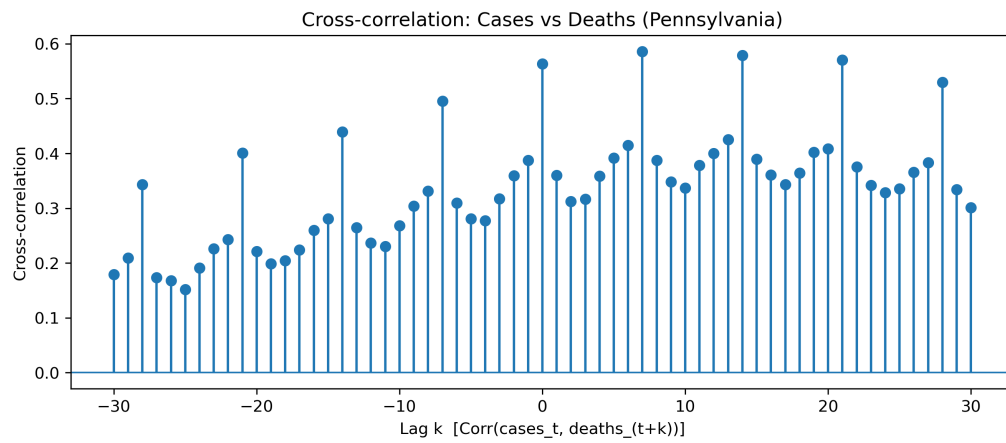


Figure 10: Cross-correlation between cases and deaths for Pennsylvania.

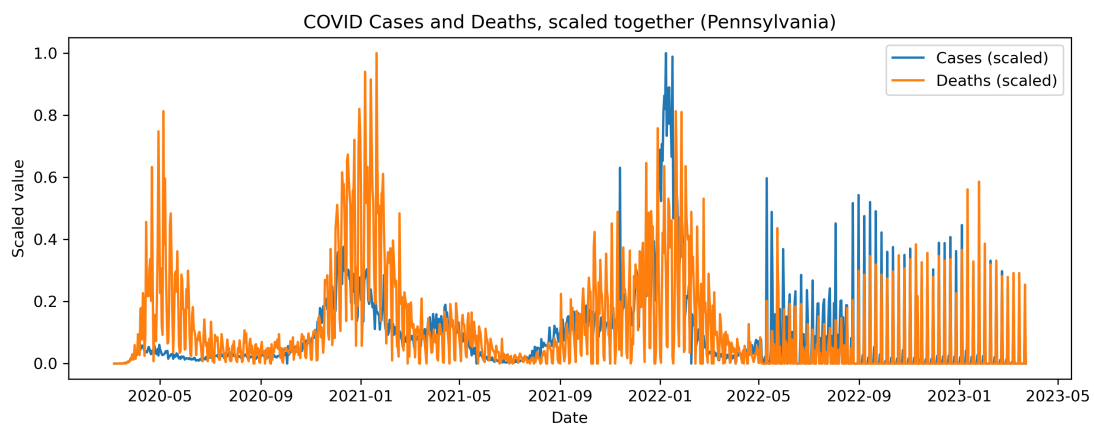


Figure 11: Scaled daily cases and deaths for Pennsylvania.

Texas

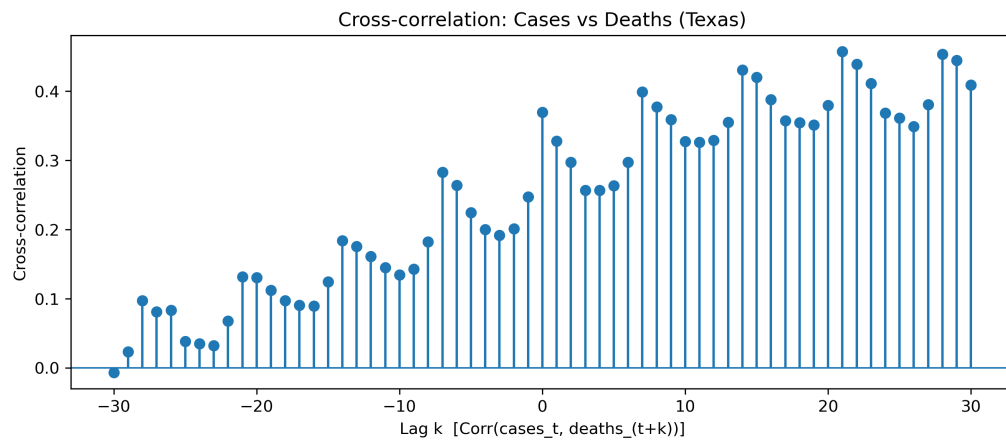


Figure 12: Cross-correlation between cases and deaths for Texas.

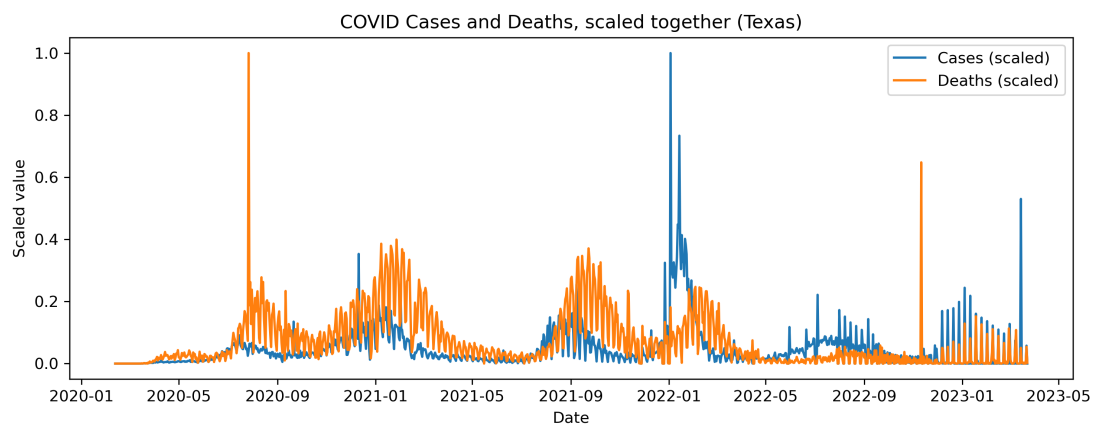


Figure 13: Scaled daily cases and deaths for Texas.